SUMAIR AHMED DERO

AI/ML Engineer | Python | Machine Learning | NLP | Computer Vision

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PROFESSIONAL SUMMARY

Third-year BS Artificial Intelligence student building end-to-end AI/ML systems in Python — from NLP pipelines and CNN-based image classifiers to recommendation engines and developer tooling. Combines applied machine learning (Scikit-learn, TensorFlow, PyTorch) with software engineering practice: REST APIs, automated testing, and interactive interfaces. Particular focus on low-resource language NLP and AI-assisted developer tools.

TECHNICAL SKILLS

Languages: Python, JavaScript

AI/ML: Scikit-learn, TensorFlow, PyTorch, CNN, LSTM, Logistic Regression, Random Forest, XGBoost, Collaborative Filtering

NLP: TF-IDF, Tokenization, Unicode Normalization, Rule-based Stemming, Lexicon-based Sentiment Analysis

Data: Pandas, NumPy, Plotly

Backend/APIs: Flask, FastAPI, REST APIs, WebSockets

Databases: SQLite, SQL, SQLAlchemy, Query Optimization

Tools: Git, GitHub, Pytest, AST-based Static Analysis, Streamlit

SOFT SKILLS

Self-Directed Learning · Analytical Thinking · Time Management

EXPERIENCE

Artificial Intelligence Intern — Syntecxhub *Sept 2026 – Present | Remote*

- Contributing to applied AI/ML projects as part of an early-stage AI internship.

Artificial Intelligence Engineer — CODE Alpha *Dec 2024 – Feb 2025 | Remote*

- Built IoT systems connecting physical sensors to cloud platforms for real-time monitoring, including microcontroller programming and device communication protocols.

Developer — ArchR Technologies (SAP Consulting Partner) *Dec 2024 – Jan 2025 | Remote*

- Built end-to-end ML models from raw data collection to prediction using Python and Scikit-learn; automated data pipelines via web scraping and Google Forms with data cleaning and feature engineering.

Research Assistant — ORIC / BBSHRRDB, University of Sindh *Apr 2026 – Present*

- Supporting applied AI-driven research methodology for institutional projects; assisting with data analysis and documentation.

SELECTED PROJECTS

Sindhi NLP Toolkit — *Python, NLP*
github.com/SUMAIRDERO7/Sindhi-nlp-toolkit-offline

- Built an offline NLP toolkit for Sindhi (Perso-Arabic script), implementing Unicode normalization, tokenization, stopword removal, and rule-based stemming.
- Implemented lexicon-based sentiment analysis and text statistics for a language with minimal existing NLP tooling.

AI-Powered Code Review Bot — *Python, AST, pytest, Streamlit*
github.com/SUMAIRDERO7/-AI-Powered-Code-Review-Bot

- Built a static-analysis tool using AST parsing to compute cyclomatic complexity, detect dead code, and generate quality scores.
- Delivered results via HTML reports and a Streamlit dashboard, with pytest coverage for core detection logic.

Crop Disease Detection (CNN) — *Python, PyTorch, Streamlit*
github.com/SUMAIRDERO7/Crop-Disease-Detection-CNN

- Built a CNN-based image classifier in PyTorch to detect disease in wheat and cotton crops.
- Deployed a Streamlit interface for image upload and prediction, with automated tests for the detection pipeline.

Product Recommendation Engine — *Python, Scikit-learn, Flask*
github.com/SUMAIRDERO7/Product-Recommendation-Engine

- Built a recommendation system combining collaborative filtering and item-similarity methods.
- Exposed the model via a Flask REST API, with a popularity-based fallback for cold-start cases.

SQL Query Optimizer — *Python, SQLite*
github.com/SUMAIRDERO7/sql-query-optimizer

- Built a query analysis tool using EXPLAIN QUERY PLAN to detect anti-patterns and recommend indexes.
- Benchmarked query performance before and after optimization.

EDUCATION

BS Artificial Intelligence — University of Sindh, Jamshoro | 2024 – 2028 (3rd year)

CERTIFICATIONS

Google AI Essentials · Google IT Automation with Python · Google Cybersecurity